Learning in the Age of AI

Lecture 01: Pedagogic Concept

Prof. Dr. Michael Granitzer

18 March 2026

Why This Teaching Style

Guiding Question

If AI can already generate summaries, definitions, and polished explanations, what is the job of the lecture?

$$\text{Lecture value} \neq \text{information transfer alone}$$

Why Might This Be True?

- AI compresses and restates efficiently
- students still struggle with meaning, transfer, and judgment
- explanation alone does not guarantee interpretation

Interpretation

The lecture should create tasks that are hard to outsource:

- interpreting formal statements
- critiquing polished answers
- comparing alternative explanations
- exposing hidden assumptions

> The lecture is valuable when it organizes thinking, not when it merely restates information.

The Question-First Module

Guiding Question

Why begin with a difficult-looking result rather than a friendly explanation?

$$\text{Question} \rightarrow \text{Tension} \rightarrow \text{Interpretation} \rightarrow \text{Stabilization}$$

Why Might This Be True?

- questions create attention
- compact results create productive uncertainty
- students must articulate meaning before being told
- synthesis stabilizes what should remain

Interpretation

The first slide in a module should not close the issue. It should open it while already showing the shape of the answer.

> Good module design gives students something to resolve, not only something to receive.

Math First, Then Meaning

Guiding Question

Why show a formal definition, equation, or compact claim before the explanation?

$$\text{formal statement first} \Rightarrow \text{students must interpret}$$

Why Might This Be True?

- it interrupts passive agreement
- it reveals whether students can assign meaning
- it surfaces hidden assumptions

Interpretation

One useful teaching move is: *"This could be an AI answer. What does it actually mean, and what is missing?"*

> Formalism becomes educational when students have to explain it in conceptual language.